CMAXX WIFI SOLUTIONS | PRE-LAUNCH WEBSITE IMPLEMENTATION BRIEF

Structured Cabling & Network Infrastructure

Complete 10/10 service-page restructure, final website copy, SEO direction, internal-linking map, technical education and FAQs

PRIMARY GOAL	CUSTOMER GOAL	CONVERSION GOAL
Rank for structured/network cabling searches	**Explain the service before the first call**	**Drive site-survey and quotation enquiries**

Prepared for CMAXX WiFi Solutions website development
27 August 2026

Developer Instruction: Replace the Current Page, Do Not Patch It

The current Infrastructure & Cabling page should be treated as a draft that requires a full content replacement. The existing layout can be reused selectively, but the page must be repositioned around structured cabling and network infrastructure, with substantial customer education, stronger local SEO relevance, clearer conversion pathways, and contextual links to CMAXX's related services.

> **CRITICAL PRE-LAUNCH CORRECTION**
>
> Remove all copy that refers to guest WiFi, "why proper CCTV matters," generic WiFi troubleshooting content, unsupported percentage claims, and any animated statistics that render as zero in the underlying HTML. The current staging page is publicly crawlable, so ensure staging remains noindex/password-protected until the final domain launches.

What the finished page must achieve

- Clearly explain what structured cabling is and why it matters to homes and businesses.
- Educate clients on Cat5, Cat5e, Cat6 and Cat6A without overwhelming them.
- Explain CCA versus 100% copper and solid versus stranded copper in plain language.
- Show that cable category alone does not guarantee a quality installation.
- Explain PoE, data points, patch panels, network cabinets, fibre backbones, outdoor cable and testing.
- Position CMAXX as a network-infrastructure specialist, not simply a contractor that "pulls cables."
- Target strong commercial-intent searches across Midrand, Johannesburg, Sandton, Centurion, Pretoria and Gauteng.
- Provide enough FAQ content to answer common objections and technical questions before a prospect calls.
- Create logical internal links to WiFi, Managed WiFi, CCTV, Smart Home, Public & Guest WiFi, WiFi assessments, Industries, Areas We Serve and Case Studies.
- Convert visitors into site-survey, quotation, call and WhatsApp enquiries.

Recommended Page Identity & SEO Setup

Element	Recommended implementation
Navigation label	Structured Cabling
Page name	Structured Cabling & Network Infrastructure
Recommended slug	/structured-cabling/
H1	Structured Cabling & Network Infrastructure in Gauteng
SEO title	Structured Cabling & Network Infrastructure Gauteng | CMAXX

Element	Recommended implementation
Meta description	Professional Cat6 and Cat6A structured cabling, data points, PoE cabling, network racks and fibre in Midrand and across Gauteng. Book a site survey.
Breadcrumb	Home > Services > Structured Cabling & Network Infrastructure
Primary CTA	Book a Cabling Site Survey
Secondary CTA	Request a Quote / WhatsApp CMAXX
Primary search theme	Structured cabling Gauteng / network cabling Gauteng
Strong secondary themes	Cat6 cabling installation; Cat6A cabling; Ethernet cabling; network cabling Midrand; data point installation; office network cabling; server rack installation; PoE cabling; fibre cabling

DEVELOPER IMPLEMENTATION NOTE

Because the site has not yet launched on its final domain, this is the ideal time to change the slug. If /infrastructure-cabling/ will exist on the live domain at any point, implement a 301 redirect to /structured-cabling/. The final page must self-canonicalise to https://www.cmaxxwifisolutions.co.za/structured-cabling/ and must not inherit a staging-domain canonical or noindex tag.

Recommended page flow

1. Hero + trust signals + primary CTA
2. What structured cabling is
3. Why the physical network layer matters
4. Structured cabling services CMAXX provides
5. Cat5 / Cat5e / Cat6 / Cat6A education
6. CCA versus 100% copper
7. Solid versus stranded copper
8. UTP versus shielded cabling
9. Indoor versus outdoor cabling
10. Ethernet distance and fibre backbones
11. PoE-ready infrastructure
12. Data points, terminations and patch panels
13. Network cabinets and server racks
14. Existing-network upgrades and fault finding
15. Testing, labelling and documentation
16. Residential and commercial use cases
17. CMAXX installation process
18. Why choose CMAXX
19. Gauteng service area
20. Related services / internal links
21. FAQs
22. Final conversion CTA

02 | FINAL CUSTOMER-FACING WEBSITE COPY

The following content is written as the recommended live-page copy. Headings are labelled by level so the developer can implement a clean HTML hierarchy.

CUSTOMER-FACING COPY

H1 - Structured Cabling & Network Infrastructure in Gauteng

A reliable network starts with the cabling behind it. CMAXX WiFi Solutions designs, installs, upgrades and tests professional structured cabling for homes, offices, commercial properties and multi-building environments across Midrand, Johannesburg, Pretoria and Gauteng.

From Cat6 and Cat6A Ethernet cabling to data points, PoE infrastructure, network cabinets, fibre backbones and cabling for WiFi access points, CCTV cameras, VoIP phones, intercoms and smart devices, we build the physical network foundation your technology depends on.

Whether you are fitting out a new office, improving an existing home network, adding wired access points, upgrading old cabling or cleaning up an unreliable server cabinet, CMAXX provides a structured solution designed for performance, maintainability and future growth.

> **HERO TRUST STRIP - RECOMMENDED**
>
> Cat6 & Cat6A | Quality copper cabling | PoE-ready infrastructure | Data points & patch panels | Network racks | Fibre backbones | Residential & commercial | Gauteng

> **DEVELOPER IMPLEMENTATION NOTE**
>
> Use one H1 only. Keep “Structured Cabling & Network Infrastructure in Gauteng” visually dominant. CTA buttons: “Book a Cabling Site Survey” and “WhatsApp CMAXX”. Avoid unsupported animated counters. If trust metrics are used, only publish verified figures.

CUSTOMER-FACING COPY

H2 - What Is Structured Cabling?

Structured cabling is the organised system of Ethernet and fibre cabling, data outlets, patch panels, network cabinets and related components that connects the devices and systems throughout a property. Instead of running random cables from one device to another, a structured installation creates a planned, labelled and maintainable network foundation.

In a modern property, that infrastructure can connect computers, printers, WiFi access points, CCTV cameras, VoIP phones, intercoms, smart TVs, servers, network switches, smart-home controllers and many other IP-based devices.

Good structured cabling is not only about achieving speed today. It should also make future changes easier, support Power over Ethernet where required, simplify troubleshooting, reduce unnecessary cable clutter and give the network room to grow.

CUSTOMER-FACING COPY

H2 - Why the Cabling Behind Your Network Matters

A fast fibre line and a high-performance router cannot compensate for poor physical infrastructure. Damaged cable, low-quality conductors, poor terminations, excessive cable lengths, incorrect connectors, badly organised cabinets and unsuitable cable routes can all create instability, reduced link speeds or intermittent faults.

CMAXX looks at the network as a complete system. We consider the internet connection, router or firewall, switches, cable routes, cable category, PoE requirements, access points, cameras, end devices and future expansion before recommending the right infrastructure.

> **KEY MESSAGE**
> The network is only as reliable as the physical path carrying the data. CMAXX builds that foundation properly.

CUSTOMER-FACING COPY

H2 - Structured Cabling Services We Provide

CMAXX provides structured cabling and network-infrastructure services for both new installations and existing properties.

Service	What it means for the client
Cat6 & Cat6A Ethernet Cabling	New permanent cable runs for homes, offices, commercial spaces, WiFi access points, CCTV cameras, workstations and other network devices.
Data Point Installation	Professional network outlets and Ethernet points terminated at the required rooms, desks, TVs, access points, cameras or equipment locations.
Patch Panels & Termination	Organised termination into keystone modules, wall outlets and patch panels so network runs can be identified, maintained and changed cleanly.
Network Cabinets & Server Racks	Installation and organisation of network racks and wall-mount cabinets for routers, firewalls, switches, patch panels, power distribution and related equipment.
PoE Network Infrastructure	Cabling designed for compatible Power over Ethernet devices such as WiFi access points, CCTV cameras, VoIP phones, intercoms and selected IoT equipment.
Fibre Backbone Cabling	Fibre links for longer distances, building-to-building connectivity, high-capacity backbones and environments where copper Ethernet is not the right fit.
Existing Cabling Upgrades	Assessment and replacement of damaged, poorly installed, outdated or unsuitable cabling, including legacy Cat5/Cat5e or questionable CCA installations where an upgrade is justified.
Network Cabling Fault Finding	Tracing unknown cable runs, checking terminations, identifying link faults, testing continuity, locating damaged connections and improving inherited network infrastructure.
Cable Management & Rack Clean-ups	Reorganising messy cabinets, relabelling runs, replacing unsuitable patch leads, improving cable routing and making future support easier.
WiFi / CCTV / Smart-Home Cabling	Purpose-planned structured cabling for access points, IP cameras, intercoms, automation controllers and connected-home equipment.

CUSTOMER-FACING COPY

H2 - Cat5, Cat5e, Cat6 and Cat6A: What Is the Difference?

Ethernet cables may look similar from the outside, but their performance categories are not the same. Choosing the right category depends on the required speed, cable length, equipment, PoE needs, environment and how long the infrastructure is expected to remain in service.

Cable category	Typical capability	CMAXX guidance
Cat5	Legacy Category cabling. Commonly associated with older 10/100 Mbps networks.	Obsolete for new professional installations. Existing Cat5 should be assessed before relying on it for a modern network.
Cat5e	Supports Gigabit Ethernet over a standard channel up to 100 m when correctly installed. Rated to 100 MHz.	Still common in existing properties and may remain suitable for many Gigabit links, but it is not our first choice for most new long-life installations.
Cat6	Supports 1 Gbps over a standard 100 m channel. Rated to 250 MHz. 10 Gigabit performance is distance- and environment-dependent and is typically limited to shorter runs.	Strong default choice for many homes, SMEs, WiFi access points, CCTV and general network installations.
Cat6A	Designed for 10 Gigabit Ethernet over a full 100 m channel. Rated to 500 MHz.	Recommended for higher-performance, enterprise, future-focused and 10GbE-ready installations where the budget and environment justify it.

For 10 Gigabit Ethernet, Cat6A is the predictable 100-metre solution. Cat6 may support 10GbE on shorter channels, but the achievable distance depends on factors such as alien crosstalk and the quality of the installed system. That is why CMAXX selects cable based on the actual application rather than simply choosing the highest category number on the box.

> **CUSTOMER EDUCATION POINT**
>
> A cable labelled “Cat6” is not automatically a good Cat6 installation. Conductor material, connector quality, cable length, routing, bend radius, terminations and testing all matter.

CUSTOMER-FACING COPY

H2 - CCA Cable vs 100% Copper Ethernet Cable

One of the most important differences customers rarely see is the material inside the cable. Some low-cost network cable is made from Copper-Clad Aluminium (CCA): an aluminium conductor with a thin copper coating. It can look similar to copper cable from the outside, but it is not equivalent to standards-compliant copper Category cabling.

CCA has higher electrical resistance than copper, which becomes especially important when the same cable is carrying power for PoE devices. It can also be less mechanically robust at terminations and may create reliability problems that are difficult to diagnose after the cable has already been installed inside ceilings, walls or conduit.

Feature	100% copper Category cable	CCA cable
Conductor	Copper conductor appropriate to recognised Category cabling systems.	Aluminium core with copper coating.
Electrical resistance	Lower resistance and better suited to data plus PoE when correctly specified.	Higher resistance; less desirable for PoE and longer permanent runs.
Permanent installation	Preferred for professional structured cabling.	Should not be treated as equivalent to compliant copper Category cable.
Long-term confidence	Better choice for infrastructure intended to remain in service for years.	Low purchase price can create hidden performance and maintenance risk.

For permanent structured cabling, CMAXX prioritises quality copper cabling selected for the required Category, environment and application. We would rather install the right foundation once than save on cable and create an expensive fault inside the building later.

> **DEVELOPER IMPLEMENTATION NOTE**
>
> Before publishing the final sentence as an absolute policy, CMAXX should operationally confirm that all permanent Category cable purchased for future projects is genuine copper and not CCA. The current recommendation is to make this the company standard.

CUSTOMER-FACING COPY

H2 - Solid Copper vs Stranded Copper: They Are Not the Same Thing

“Solid copper” and “stranded copper” describe how a copper conductor is constructed. This is different from CCA, which refers to the conductor material itself.

Type	Best suited to	Why
Solid copper conductor	Permanent horizontal cabling inside walls, ceilings, ducts and structured cable routes.	Commonly used for fixed runs because it provides a robust permanent link and lower conductor resistance for the same gauge.
Stranded copper conductor	Flexible patch leads between a wall outlet or patch panel and network equipment.	More flexible and better suited to short cords that are moved, bent or handled repeatedly.
CCA	Not CMAXX’s recommended substitute for professional permanent Category cabling.	It is aluminium-based rather than an all-copper conductor and should not be confused with stranded copper.

In a professionally structured installation, permanent building runs are normally terminated to a patch panel or data outlet, and flexible patch cords are then used to connect equipment. This makes the network easier to service and reduces unnecessary movement of fixed cabling.

CUSTOMER-FACING COPY

H2 - UTP vs Shielded Ethernet Cabling

Most standard office and residential networks use unshielded twisted-pair cabling, often written as U/UTP. In environments with higher levels of electrical noise or specific technical requirements, shielded designs such as F/UTP may be more appropriate.

Shielding is not a simple “better cable” upgrade. A shielded cabling system must be selected and installed correctly, including compatible components and the required bonding or grounding approach. CMAXX assesses the environment before deciding whether standard unshielded cabling or a shielded solution is appropriate.

> **SIMPLE RULE**
> Use the cable construction that suits the environment. Do not pay for shielding that is not required, and do not install shielded cable incorrectly.

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H2 - Indoor vs Outdoor Ethernet Cabling

Indoor Ethernet cable is not automatically suitable for exposed outdoor use. Sunlight, temperature changes, moisture and physical exposure can degrade unsuitable cable jackets over time.

For outdoor, rooftop, perimeter or between-building routes, CMAXX selects cable and protection appropriate to the environment. Depending on the installation, this may include UV-resistant outdoor-rated Ethernet cable, protective conduit, suitable glands and weather-aware routing. Where distances or environmental conditions make copper impractical, fibre may be the better backbone.

We also plan routes to reduce unnecessary exposure, sharp bends, water entry points and avoidable maintenance problems.

CUSTOMER-FACING COPY

H2 - How Far Can Ethernet Cabling Run?

For standard twisted-pair Ethernet structured cabling, the normal design limit is a 100-metre channel. In a conventional structured-cabling layout, the permanent installed link is designed around a maximum of 90 metres, leaving the remaining allowance for patch cords and connection hardware within the full channel.

If a required run is too long, the answer is not to keep extending copper cable and hope for the best. CMAXX can redesign the link using an intermediate network location or fibre backbone, depending on the property and application.

> **IMPORTANT**
> Longer is not better. Cable length, insertion loss, PoE voltage drop and the end-to-end application all need to be considered together.

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H2 - PoE-Ready Cabling for WiFi, CCTV, VoIP and Intercoms

Power over Ethernet (PoE) allows compatible devices to receive network data and electrical power through the Ethernet connection. This can simplify installations by reducing the need for a separate power point at every device location.

Common PoE devices include ceiling-mounted WiFi access points, IP CCTV cameras, VoIP phones, video intercoms, access-control devices and selected smart-building equipment.

A PoE-ready design is more than connecting a PoE switch. CMAXX considers cable quality, conductor size, cable length, power requirements, switch capacity, heat in large cable bundles and the device's PoE standard before finalising the infrastructure.

> **INTERNAL LINK IN THIS SECTION**
> Link "WiFi access points" to Business & Office WiFi Solutions and "IP CCTV cameras" to CCTV & Security Solutions.

CUSTOMER-FACING COPY

H2 - Data Points, RJ45 Outlets, Patch Panels and Termination

A structured network should be easy to identify and maintain. Instead of leaving permanent cables loose in a cabinet or hanging from a wall, CMAXX can terminate fixed cabling into labelled data outlets, keystone modules and patch panels where appropriate.

Typical data-point locations include workstations, home offices, TVs, printers, WiFi access points, CCTV camera positions, intercoms, server rooms and other fixed network devices.

Correct termination helps maintain cable performance, simplifies future moves and changes, and gives a technician a clear path from a room outlet back to the network cabinet.

CUSTOMER-FACING COPY

H2 - Network Cabinets & Server Rack Installations

The network cabinet is the central point where structured cabling and active network equipment come together. A poorly organised cabinet can turn a simple fault into hours of unnecessary troubleshooting.

CMAXX designs, installs and reorganises network cabinets for homes and businesses. Depending on the project, a cabinet may include a router or firewall, network switches, PoE switches, patch panels, cable managers, power distribution, UPS equipment, fibre termination and spare space for future expansion.

We focus on clean routing, sensible labelling, ventilation, accessibility and serviceability so the cabinet remains understandable long after the installation is complete.

> **CONVERSION MICRO-CTA**
> Inherited a messy cabinet? Ask CMAXX for a network rack clean-up and documentation assessment.

CUSTOMER-FACING COPY

H2 - Fibre Cabling for Backbones and Longer Distances

Copper Ethernet is excellent for normal access runs, but it is not the answer to every network link. Fibre is often the better choice for longer distances, high-capacity backbones or connections between buildings.

Single-mode fibre is commonly selected for longer-distance and scalable backbone links, while multimode fibre can be appropriate for shorter high-speed links depending on the transceivers, distance and system design. CMAXX assesses the application before recommending the fibre type and active equipment required at each end.

Typical fibre use cases include office backbones, warehouses, farms, campuses, estates, outbuildings and properties where network links need to extend beyond normal copper Ethernet distances.

CUSTOMER-FACING COPY

H2 - Upgrading Old or Problematic Network Cabling

Not every existing cable needs to be replaced simply because a newer category exists. The correct approach is to assess what is installed, what performance is required and whether the current infrastructure is actually creating a bottleneck or reliability problem.

CMAXX can inspect inherited cabling, trace unknown runs, identify legacy or damaged cable, check suspicious terminations, review link speeds and recommend targeted upgrades. This is particularly useful when a property has changed owners, undergone renovations or accumulated years of undocumented network additions.

Where Cat5e remains healthy and meets the application, it may continue to serve the network. Where old, damaged, unsuitable or low-quality cable is limiting the system, CMAXX can replace the affected runs with an appropriate modern solution.

CUSTOMER-FACING COPY

H2 - Cable Testing, Verification, Labelling & Documentation

A cable is not finished when it has simply been pulled through the ceiling. Each completed run should be terminated correctly, identified and checked before handover.

Depending on the project scope, CMAXX can perform wire-map and continuity verification, check terminations and link status, validate network connectivity, label data points and patch-panel positions, and document the installed routes or port schedule. Where a project requires formal standards-based cable certification, the testing method and reporting requirement should be specified separately before work begins.

> **TERMINOLOGY RULE FOR THE WEBSITE**
>
> Use “tested and verified” as the normal claim. Only use “certified” when a project is actually tested with appropriate certification equipment against the relevant cabling standard and a certification report is issued.

CUSTOMER-FACING COPY

H2 - Structured Cabling for Homes, Offices and Commercial Properties

CMAXX supports structured network infrastructure across a wide range of environments:

- Homes and residential estates - home offices, smart TVs, WiFi access points, CCTV, intercoms and smart-home infrastructure.
- Offices and SMEs - workstations, printers, access points, IP phones, meeting rooms, server rooms and network cabinets.
- Student accommodation and multi-tenant properties - scalable access-point, CCTV and building-network infrastructure.
- Retail, hospitality and venues - staff networks, guest WiFi infrastructure, POS connectivity, cameras and operational systems.
- Schools, churches and community facilities - reliable wired backbones for WiFi, administration, cameras and shared spaces.
- Warehouses, farms and multi-building properties - long runs, outdoor infrastructure, fibre or point-to-point integration and distributed access points.

- New builds, renovations and office fit-outs - planned network points and cable routes before walls and ceilings are closed.

CUSTOMER-FACING COPY

H2 - Our Structured Cabling Process

Stage	What CMAXX does
01 - Assess	We inspect the property, network requirements, existing cabinet, cable routes, equipment locations, PoE needs and future expansion.
02 - Design	We decide where data points, patch panels, switches, cabinets, access points and fibre links should be positioned.
03 - Specify	We select the appropriate cable category, conductor type, indoor/outdoor construction, shielding and supporting hardware.
04 - Install	Cabling is routed, secured and terminated with attention to neatness, accessibility and minimal disruption.
05 - Test & Label	Runs are checked, labelled and mapped so the network can be maintained logically.
06 - Commission & Handover	Connected devices and network paths are verified, documentation is handed over where included, and future support options are discussed.

CUSTOMER-FACING COPY

H2 - Why Choose CMAXX for Network Cabling?

Structured cabling is most valuable when it is designed by people who understand the network that will run on top of it. CMAXX combines physical cabling with practical experience in WiFi, routing, switching, PoE, CCTV, smart-home connectivity and managed networks.

- Network-first design - cabling is planned around the devices, traffic and coverage requirements it must support.
- Quality-focused cable selection - the category and conductor are chosen for the application rather than price alone.
- Clean, maintainable installations - labelling, patching and cabinet organisation are treated as part of the job.
- Residential and commercial capability - from one additional home data point to multi-floor or multi-building infrastructure.
- Integrated service capability - one team can coordinate cabling with WiFi, CCTV, network switching, smart-home and managed-service requirements.
- Local Gauteng support - CMAXX is based in Midrand and supports customers across major Gauteng business and residential areas.
- Business procurement readiness - CMAXX is a B-BBEE Level 1 contributor and COIDA registered.

CUSTOMER-FACING COPY

H2 - Structured Cabling Across Midrand, Johannesburg & Gauteng

CMAXX WiFi Solutions provides structured cabling and network infrastructure services from Midrand across Johannesburg, Sandton, Centurion, Pretoria, Randburg, Fourways, Bryanston, Waterfall, Kyalami, Bedfordview, Edenvale, Kempton Park and surrounding Gauteng areas.

If your property is outside these locations, contact us with the site address and project scope. Larger or multi-site projects can be assessed based on location and requirements.

> INTERNAL LINK
>
> Link “surrounding Gauteng areas” to Areas We Serve. Do not repeat a giant suburb list merely to target keywords; the dedicated Areas We Serve page should carry the full location coverage.

CUSTOMER-FACING COPY

H2 - Related Network & Connectivity Services

Structured cabling is often one part of a larger connectivity project. CMAXX can also help with:

Internal-link anchor	Context
WiFi & Network Solutions	Overall network design, routers, switches, access points and connectivity troubleshooting.
Home WiFi Solutions	Whole-home coverage using correctly placed access points, wired backhaul or suitable mesh designs.
Business & Office WiFi Solutions	Secure, scalable office WiFi with wired access points, VLANs, switching and capacity planning.
WiFi Assessment & Troubleshooting	Diagnose speed, coverage, roaming, cabling and configuration problems before replacing equipment.
Outdoor WiFi Solutions	Coverage for gardens, yards, entertainment areas, campuses, warehouses and outdoor operational spaces.
Managed WiFi & Networks	Ongoing monitoring, support, maintenance and network management for business clients.
CCTV & Security Solutions	IP camera systems and network infrastructure for surveillance and remote viewing.
Smart Home Solutions	Reliable network infrastructure for connected lighting, security, access, entertainment and automation.
Public & Guest WiFi	Guest networks, captive portals and managed connectivity for venues and public-facing businesses.

> DEVELOPER IMPLEMENTATION NOTE
>
> Use real contextual <a href> links, not JavaScript-only click handlers. Keep anchor text descriptive and close to the phrases above so related pages are crawlable and meaningful to users and search engines.

H2 - Structured Cabling Questions Answered

Use an accessible accordion on mobile if desired, but the question and answer text must remain present in the rendered HTML/DOM and crawlable. Do not rely on FAQ schema for rich-result visibility; Google generally limits FAQ rich results to authoritative government and health sites.

What is structured cabling?

Structured cabling is an organised system of Ethernet or fibre cables, data outlets, patch panels, network cabinets and related components that connects devices throughout a building. It provides a cleaner, easier-to-maintain foundation than ad-hoc cables running directly between individual devices.

What is the difference between Cat5, Cat5e, Cat6 and Cat6A?

Cat5 is legacy cabling and is not recommended for new installations. Cat5e commonly supports Gigabit Ethernet up to a standard 100-metre channel. Cat6 provides improved performance and supports 1 Gigabit over 100 metres, while 10 Gigabit use is generally limited to shorter channels. Cat6A is designed for 10 Gigabit Ethernet over the full 100-metre channel.

Should I install Cat6 or Cat6A?

For many homes, SMEs, CCTV and WiFi access-point installations, Cat6 is an excellent choice. Cat6A is worth considering when you need 10GbE over longer runs, higher-performance enterprise infrastructure, additional future headroom or a long-life installation where the extra cable size and cost are justified. CMAXX recommends the category after assessing the application.

Does CMAXX recommend Cat5e for new installations?

Cat5e can still provide reliable Gigabit connectivity when it is correctly installed and remains common in existing buildings. For many new permanent installations, however, CMAXX will normally consider Cat6 or Cat6A first because the infrastructure is expected to remain in place for years.

What is CCA Ethernet cable?

CCA stands for Copper-Clad Aluminium. The conductor has an aluminium core with a copper coating. It is cheaper than all-copper Category cable but has higher resistance and should not be treated as equivalent to standards-compliant copper structured cabling, particularly for long-life or PoE-heavy installations.

Does CMAXX use CCA cable?

For permanent professional structured cabling, CMAXX prioritises quality copper cabling that is appropriate to the required Category and application. CCA is not our recommended substitute for proper copper Category cable.

What is the difference between solid copper and stranded copper cable?

Solid-conductor copper cable is commonly used for fixed permanent building runs because it is suited to structured horizontal cabling. Stranded copper is more flexible and is commonly used for patch leads that connect equipment to wall outlets or patch panels. Both can be copper; CCA is a different conductor material altogether.

How long can an Ethernet cable run?

A standard twisted-pair Ethernet channel is normally designed around a maximum of 100 metres. A conventional permanent installed link is typically limited to 90 metres, leaving the remaining channel allowance for patch cords and connection hardware. For longer links, fibre or a different network design may be required.

Can Ethernet cabling power WiFi access points and CCTV cameras?

Yes. Compatible Power over Ethernet equipment can carry network data and electrical power over the Ethernet connection. PoE is commonly used for WiFi access points, IP cameras, VoIP phones and intercoms. The cable, switch, distance and device power requirement all need to be designed correctly.

Can you install Ethernet cable next to electrical power cables?

Data cabling should be routed with appropriate separation from mains electrical cabling and in accordance with applicable installation rules and the equipment and cabling specifications. CMAXX plans cable routes to reduce electrical interference and avoid unsafe or non-compliant shared pathways.

What is the difference between UTP and shielded cable?

Unshielded twisted-pair cable is suitable for many normal residential and office environments. Shielded cabling can be useful in electrically noisy environments or where a particular design requires it, but the shielding system must be implemented correctly. CMAXX selects the appropriate construction after assessing the site.

Do I need outdoor-rated Ethernet cable outside?

Yes, exposed outdoor cable should be suitable for the environmental conditions. CMAXX can use outdoor or UV-resistant cable, conduit and appropriate routing where required. Standard indoor cable should not simply be left exposed to sunlight and weather.

Can CMAXX add data points to an existing home or office?

Yes. We assess available ceiling spaces, conduits, risers, wall routes and cabinet locations and then plan the cleanest practical route. Retrofit work is designed to minimise disruption while still producing a maintainable installation.

Can you upgrade old Cat5 or Cat5e cabling?

Yes. We can inspect the existing network and determine whether the older cabling is actually limiting performance or has faults. Healthy Cat5e may still be useful for some Gigabit links, while damaged, unsuitable or legacy runs can be replaced with Cat6, Cat6A or fibre where appropriate.

Can you install cabling for WiFi access points?

Yes. Wired access points are one of the most common uses for structured cabling. CMAXX can design the WiFi coverage, choose access-point locations, install the Ethernet runs and PoE switching, and configure the network as one coordinated solution.

Can you install network cabling for CCTV cameras?

Yes. CMAXX installs network infrastructure for compatible IP CCTV systems, including PoE camera cabling, switches, cabinet integration and the connectivity required for recording and remote viewing.

Do you install network cabinets and server racks?

Yes. We install and organise wall-mount network cabinets and larger racks depending on the project. We can integrate patch panels, routers, switches, PoE switches, cable management, power distribution and suitable space for future expansion.

Can you clean up an existing messy server or network cabinet?

Yes. CMAXX can trace and relabel connections, reorganise patch leads, improve cable management, replace unsuitable components where required and document the cabinet so future maintenance becomes easier.

Do you test the cables after installation?

Yes. Installed runs should be checked and verified before handover. The exact test level depends on the project scope. CMAXX can perform connectivity, wire-map and continuity checks and document labelling. If formal standards-based certification reports are required, that requirement should be agreed before the project starts.

What is the difference between cable testing and cable certification?

Basic testing confirms items such as wire map, continuity or whether a network link is functioning. Formal cable certification uses specialist certification equipment to measure the installed link against a defined cabling standard and produces a pass/fail report. The two should not be described as the same service.

When should fibre be used instead of Ethernet copper cable?

Fibre is often the better choice when the link needs to exceed normal copper Ethernet distances, connect separate buildings, provide a high-capacity backbone or meet a specific environmental or scalability requirement. CMAXX can recommend single-mode or multimode fibre based on the actual link design.

How long does a structured cabling installation take?

The timeline depends on the number of points, building access, ceiling type, conduit availability, drilling or trunking requirements, cabinet work and whether the installation is residential or commercial. Small additions may be completed quickly, while office fit-outs or multi-building projects require a planned installation schedule.

Can cabling be hidden so the installation looks neat?

Where the building allows it, cables can be routed through ceiling spaces, conduits, ducts, risers or suitable wall routes. Where concealed routes are not practical, neat trunking or protected surface routes can be considered. The site survey determines the best balance between appearance, serviceability and cost.

Do you provide structured cabling for both homes and businesses?

Yes. CMAXX supports homes, offices, SMEs, student accommodation, retail, hospitality, schools, churches, warehouses, farms and other properties across Gauteng. The design and hardware are matched to the size and operational needs of each site.

Which areas do you provide network cabling services in?

CMAXX is based in Midrand and supports clients across Johannesburg, Sandton, Centurion, Pretoria, Randburg, Fourways, Bryanston, Waterfall, Kyalami, Bedfordview, Edenvale, Kempton Park and surrounding Gauteng areas. Contact us for larger projects outside the listed areas.

Can you assess an existing network before I decide what to replace?

Yes. In many cases an assessment is the best first step. CMAXX can review the existing cabling, cabinet, switches, WiFi and connected devices, identify the actual bottlenecks and then recommend only the changes that are justified.

> **FAQ CTA**
>
> Still unsure which cable or network design your property needs? Book a CMAXX cabling site survey and we will assess the existing infrastructure, required devices, cable routes and future requirements before recommending a solution.

04 | FINAL CONVERSION SECTION

Final CTA - customer-facing copy

H2 - Build the Network Foundation Properly the First Time

Whether you need one additional Ethernet point or a complete structured cabling system for an office, home, warehouse, student property or multi-building site, CMAXX can assess the requirement and design the right infrastructure around it.

Tell us what you need to connect, where the equipment will be located and what is not working today. We will help you determine the right cable category, routes, data points, PoE requirements, network cabinet and backbone design for the property.

> **BUTTONS**
>
> PRIMARY: Book a Cabling Site Survey | SECONDARY: Request a Quote | TERTIARY: WhatsApp CMAXX / Call Us

05 | INTERNAL LINKING MAP

Required Internal Links From This Page

Source section / context	Recommended anchor text	Recommended target
Hero / opening	WiFi & Network Solutions	/wifi-network-solutions/
PoE section	Business & Office WiFi Solutions	/business-office-wifi-solutions/
Residential use case	Home WiFi Solutions	/home-wifi-solutions/
Outdoor cable section	Outdoor WiFi Solutions	/outdoor-wifi-solutions/
Upgrades / fault finding	WiFi Assessment & Troubleshooting	/wifi-assessment-troubleshooting/
PoE / cameras	CCTV & Security Solutions	/cctv-security-solutions/
Smart home use case	Smart Home Solutions	/smart-home/
Commercial / ongoing support	Managed WiFi & Networks	/managed-wifi/
Retail / hospitality	Public & Guest WiFi	/public-guest-wifi/
Sector use cases	Industries We Serve	/industries/
Location section	Areas We Serve	/areas-we-serve/
Trust / proof section	CMAXX Case Studies	/case-studies/ or final Case Studies slug

Required Links Back to the Structured Cabling Page

Internal linking must work both ways. The developer should add contextual links to /structured-cabling/ from the following pages whenever they mention data points, wired backhaul, PoE, Ethernet, network racks or physical infrastructure:

- WiFi & Network Solutions
- Home WiFi Solutions
- Business & Office WiFi Solutions
- WiFi Assessment & Troubleshooting
- Outdoor WiFi Solutions
- Managed WiFi
- CCTV & Security Solutions
- Smart Home Solutions
- Public & Guest WiFi
- Relevant case studies and blog articles

> **DEVELOPER IMPLEMENTATION NOTE**
>
> Recommended backlink anchor variations: “structured cabling”, “Cat6 network cabling”, “Ethernet data points”, “wired network infrastructure”, “PoE cabling”, and “network cabling installation”. Keep anchors natural and context-specific rather than repeating one exact keyword everywhere.

06 | SEO IMPLEMENTATION INSTRUCTIONS

On-Page SEO Requirements

The content must read naturally for a real customer first. Google explicitly recommends helpful, reliable, people-first content and descriptive titles/headings rather than keyword stuffing. This page is intentionally comprehensive because it answers real purchase and technical questions - not because Google has a preferred word count.

SEO element	Developer instruction
Title tag	Use the recommended unique title. Do not duplicate another service page title.
H1	Use one H1 only: “Structured Cabling & Network Infrastructure in Gauteng”.
H2/H3 hierarchy	Use H2s for major topics and H3s only for genuine subsections/cards. Do not use heading tags purely for visual styling.
Meta description	Use the recommended description or a close equivalent that includes the service, location and value proposition naturally.
Canonical	Self-canonical to the final cmaxxwifisolutions.co.za URL. Never canonical to the staging domain.
Indexing	Final production page must be indexable. Staging should remain noindex/password protected.
Internal links	Use crawlable <a href> links and descriptive anchor text. Do not hide critical links behind JavaScript-only click events.

SEO element	Developer instruction
Images	Use original CMAXX installation photos wherever possible. Compress them, lazy-load below-the-fold media and use descriptive filenames/alt text.
FAQ	Visible, helpful answers are valuable. FAQPage schema may be included if accurate, but do not expect a Google FAQ rich result for a normal commercial service page.
Schema	Consider Service and BreadcrumbList structured data, plus organisation/local-business information at the appropriate site level. Mark up only facts visible and true on the page/site.
NAP	Keep company name, telephone, email and Midrand/Gauteng service information consistent with the site and Google Business Profile.
CTAs	Include meaningful CTAs above the fold, around the middle of the page, after FAQs and at the final section.
Mobile UX	Tables should become scrollable/responsive cards where needed. FAQ accordions must remain accessible and crawlable.

Keyword & Topic Map

Priority	Search themes to cover naturally
Primary	structured cabling Gauteng; network cabling Gauteng; structured cabling Midrand; network cabling Midrand
Service intent	Cat6 cabling installation; Cat6A cabling installation; Ethernet cabling installation; data point installation; network point installation
Commercial	office network cabling; business network cabling; server rack installation; network cabinet installation; patch panel installation
Technical	CCA vs copper cable; solid copper Ethernet; PoE cabling; UTP vs shielded cable; outdoor Ethernet cable; fibre backbone cabling
Related intent	WiFi access point cabling; CCTV network cabling; smart home network cabling; network cabling testing; cable fault finding
Local modifiers	Johannesburg; Sandton; Centurion; Pretoria; Randburg; Fourways; Bryanston; Waterfall; Kyalami; Bedfordview; Edenvale; Kempton Park

DEVELOPER IMPLEMENTATION NOTE

Do not try to insert every keyword into every heading. Use the primary topic in the title/H1, then allow secondary terms to appear where they genuinely explain a service. This avoids spammy repetition and makes the page more useful.

Recommended Image & Alt-Text Plan

Image opportunity	Suggested filename	Suggested alt text
Cat6/Cat6A cable and termination	cat6-cat6a-structured-cabling-gauteng.jpg	Cat6 and Cat6A structured cabling installation by CMAXX WiFi Solutions
Patch panel / clean rack	network-rack-patch-panel-midrand.jpg	Labelled network rack and patch panel installation in Midrand
Ceiling AP cable run	poe-wifi-access-point-cabling.jpg	PoE Ethernet cabling for a ceiling WiFi access point
Data point / wall outlet	ethernet-data-point-installation.jpg	Professional Ethernet data point installation
Outdoor protected run	outdoor-network-cabling-gauteng.jpg	Protected outdoor network cabling installation in Gauteng
Fibre / backbone work	fibre-backbone-network-installation.jpg	Fibre backbone network link for a multi-building property

PHOTO PRIORITY

Use genuine CMAXX project photography ahead of generic stock images wherever possible. Real rack, cabling, access-point and data-point photos reinforce first-hand expertise and trust.

07 | TECHNICAL ACCURACY GUARDRAILS

Claims the Developer Must Not Overstate

Topic	Safe website wording	Avoid unless verified
Cable certification	“Tested and verified”; formal certification available only where specifically scoped.	“Every project certified” if no certification analyser/report is used.
Cat6 10GbE	Cat6 may support 10GbE on shorter runs; environment and alien crosstalk matter.	“Cat6 supports 10Gbps to 100 m.”
Cat6A 10GbE	Designed to support 10GbE over a 100 m channel when the system is correctly installed.	Unqualified claims that any cable marked Cat6A will always achieve 10GbE regardless of installation.
CCA	CMAXX prioritises proper copper Category cable; CCA should not be treated as equivalent to compliant copper Category cabling.	South Africa-specific “illegal” claims unless supported by applicable local code/legal advice.
Failure statistics	Explain physical-layer faults qualitatively.	“70% of network failures are cabling” without a specific defensible source.
Experience / project counts	Use only verified company figures.	Animated placeholder numbers or invented metrics.

Topic	Safe website wording	Avoid unless verified
Free assessment	Use “Book a Site Survey” unless CMAXX has confirmed the assessment is free for this service.	“Free” merely because another page offers a free WiFi assessment.

Technical Reference Notes Behind the Copy

- CommScope publishes Category 5e at 100 MHz, Category 6 at 250 MHz and Category 6A at 500 MHz, with Cat6A designed for 10GbE up to 100 m.
- Fluke Networks explains that Cat6 10GBASE-T capability is distance and alien-crosstalk dependent: approximately 37 m is the baseline, with operation between 37 m and 55 m depending on the environment; Cat6A is the 100 m 10GBASE-T solution.
- Fluke Networks documents the conventional 90 m permanent-link and 100 m channel design limits for balanced twisted-pair cabling.
- The Communications Cable & Connectivity Association (CCCA), citing industry standards, warns that CCA conductors should not be marketed as equivalent standards-compliant Category communications cable and highlights higher resistance and reliability concerns.
- Belden Category 6 premises cable examples use solid bare copper conductors, while flexible patch-cord products commonly use stranded bare copper; this supports the plain-language fixed-run versus patch-lead distinction used on the page.
- Google Search Central recommends concise descriptive titles, people-first content, crawlable internal links and descriptive anchor text. Google also notes that FAQ rich results are generally limited to authoritative government and health sites.

Source set used for technical/SEO validation

- CommScope - “What is Cat 6A and how is it the best choice for enterprise networks?”
- Fluke Networks - “10GBASE-T Field Testing Requirements” and twisted-pair 10 Gigabit cabling guidance
- Fluke Networks - “Going the Distance” / DSX CableAnalyzer length guidance
- Communications Cable & Connectivity Association (CCCA) - Copper-Clad Aluminium conductor compliance and risk resources
- Belden - Category 6 premises/horizontal cable and copper patch-cord product specifications
- Google Search Central - Helpful, reliable, people-first content; title-link guidance; link best practices; FAQ rich-result changes

08 | DESIGN & CONVERSION DIRECTION

Recommended Visual Treatment for the Live Page

- Hero: strong CMAXX project image of organised cabling, patch panel or network rack - not a generic “blue cable” stock photo.
- Use visual education cards for Cat5/Cat5e/Cat6/Cat6A and CCA vs copper. These sections should be easy to scan on mobile.
- Use icons for services, but pair every icon with meaningful text; do not let icon cards replace the detailed copy.
- Include one or two real before/after rack or cabling examples if CMAXX has suitable photos.
- Repeat a small “Book a Cabling Site Survey” CTA after the cable-education section and again before the FAQs.

- Use a responsive comparison table or stacked mobile cards for cable categories.
- FAQ accordions are acceptable for usability, provided all answer text remains in the page HTML and can be accessed without relying on non-crawlable scripts.
- Keep the page fast. Compress project photography and avoid heavy decorative animation that delays core content.
- Show phone, WhatsApp and enquiry options clearly. Do not force every visitor into one contact method.
- Place related-service cards after the process or service-area section so visitors can move deeper into the relevant CMAXX service cluster.

09 | DEVELOPER LAUNCH CHECKLIST

Pre-Launch Acceptance Checklist

☐ Replace all current guest WiFi/CCTV copy errors on the cabling page.

☐ Rename page to Structured Cabling & Network Infrastructure.

☐ Set final slug to /structured-cabling/ and add redirect if required.

☐ Use the approved title tag, meta description, H1 and breadcrumb.

☐ Ensure there is only one H1.

☐ Publish the complete educational sections, not only the service cards.

☐ Include the Cat5/Cat5e/Cat6/Cat6A comparison.

☐ Include CCA vs copper and solid vs stranded explanations.

☐ Include PoE, outdoor cabling, distance, fibre, rack and testing sections.

☐ Do not publish unsupported 70% failure statistics.

☐ Do not use “certified” as a blanket claim without the appropriate testing scope.

☐ Remove zero-value animated counters or make the underlying HTML contain verified figures.

☐ Implement all required internal links with descriptive, crawlable anchors.

☐ Add links back to this page from WiFi, CCTV, Smart Home and Managed WiFi pages.

☐ Use original CMAXX photos where available and add descriptive alt text.

☐ Implement FAQ section with accessible mobile accordion behaviour if used.

☐ Ensure staging is noindex/password protected and production is indexable.

☐ Set production canonical to the final CMAXX domain.

☐ Check mobile layout, tables, accordions and CTA buttons on common screen sizes.

☐ Verify phone, WhatsApp, email and enquiry buttons work.

☐ Check Core Web Vitals/page load after images and scripts are added.

☐ Add the URL to the XML sitemap and submit/request indexing in Google Search Console after launch.

☐ Confirm navigation, breadcrumb and footer link labels use the final page name consistently.

> **FINAL ACCEPTANCE STANDARD**
>
> The page is ready only when a customer can land on it knowing almost nothing about Ethernet cabling and leave understanding what CMAXX installs, why cable quality matters, the difference between the main cable options, how the installation process works, and exactly how to request an assessment or quote.